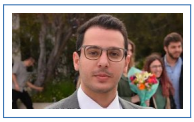
Antonios P. Sarikas

AI for Chemistry & Materials

antonios.sarikas@gmail.com
adosar.github.io/
adosar
Zv2Uk0AAAAAJ
0009-0008-8420-927X
3 January 2000



Βελτίων ἢ ἐξ ἐλαττόνων.

Research interests

Topics AI for Materials Discovery • Representation Learning • Foundation Models • Inverse Design

Education

2024–TBA PhD, Chemistry, Department of Chemistry, University of Crete
AI for porous materials

2022–2024 MSc, Physical Chemistry, Department of Chemistry, University of Crete
9.86 (Excellent)

2018–2022 BSc, Chemistry, Department of Chemistry, University of Crete
8.64 (Excellent)
First of the class – obtained in 3.5/4 years

Master thesis

Title From Potential Energy Surface to Gas Adsorption via Deep Learning

Description Developed a deep learning framework for predicting gas adsorption in porous materials using 3D energy images.

Document Click here to access it

Bachelor thesis

Title Screening MOFs for H_2 Storage via Machine Learning

Description Developed a machine learning framework for rapid indentification of top-performing MOFs for hydrogen storage.

Scholarships – Awards – Highlights

2025 **Excellence Award**, "Professor Zoe Dimitriadis Prize", University of Crete

2025 **Best MSc Thesis Award** (3rd place), Federation of European Materials Societies

2025 **Postgraduate Fellowship**, Vasileios Apollon Anagnostakis Bequest Fellowship

2025 **Best MSc Thesis Award** (1st place), Hellenic Society for the Science and Technology of Condensed Matter

2025 **Paper featured in *Nature Portfolio Nobel Prize in Chemistry 2025***. "Gas adsorption meets deep learning: voxelizing the potential energy surface of metal-organic frameworks"

2025 **Paper ranked 9th in *Scientific Reports 2024 Materials Science Top 100***. "Gas adsorption meets deep learning: voxelizing the potential energy surface of metal-organic frameworks".

2024 **Postgraduate Fellowship**, Independent Power Transmission Operator S.A. Fellowship

2024 **Postgraduate Fellowship**, Maria Michail Manassaki Bequest Fellowship

2023 **Chatzimarinaki Award**, Department of Chemistry, University of Crete

2022 **Undergraduate Scholarship**, Stamatiou Foundation

2021 **Undergraduate Scholarship**, Stamatiou Foundation

2020 **Undergraduate Scholarship**, Stamatiou Foundation

¶ Publications

[1] **Antonios P. Sarikas** and George E. Froudakis. "IntelliPore: A Foundation Model for Gas Adsorption in Porous Materials". In: (2026). DOI: 10.26434/chemrxiv.15008693/v1.

[2] Alexios P. Drosos, **Antonios P. Sarikas**, and George E. Froudakis. "Predicting hydrogen storage in MOFs: Representation matters". In: *International Journal of Hydrogen Energy* 254 (July 2026), p. 156056. DOI: 10.1016/j.ijhydene.2026.156056.

[3] **Antonios P. Sarikas**, Konstantinos Gkagkas, and George E. Froudakis. "RetNeXt: A Pretrained Model for Transfer Learning Across the MOF Adsorption Space". In: *Journal of Chemical Information and Modeling* (Feb. 2026). DOI: 10.1021/acs.jcim.5c02698.

[4] Aydin Ozcan, François-Xavier Coudert, Sven M. J. Rogge, Greta Heydenrych, Dong Fan, **Antonios P. Sarikas**, Seda Keskin, Guillaume Maurin, George E. Froudakis, Stefan Wuttke, and Ilknur Erucar. "Artificial Intelligence Paradigms for Next-Generation Metal–Organic Framework Research". In: *Journal of the American Chemical Society* (June 2025). DOI: 10.1021/jacs.5c08214.

[5] **Antonios P. Sarikas**, Konstantinos Gkagkas, and George E. Froudakis. "Gas adsorption meets geometric deep learning: points, set and match". In: *Scientific Reports* 14.1 (Nov. 2024), p. 27360. DOI: 10.1038/s41598-024-76319-8.

[6] **Antonios P. Sarikas**, Konstantinos Gkagkas, and George E. Froudakis. "Gas adsorption meets deep learning: voxelizing the potential energy surface of metal-organic frameworks". In: *Scientific Reports* 14.1 (Jan. 2024). DOI: 10.1038/s41598-023-50309-8.

[7] **Antonios P. Sarikas**, George S. Fanourgakis, Konstantinos Gkagkas, and George E. Froudakis. "Comparison of machine learning approaches for the identification of top-performing materials for hydrogen storage". In: *Sustainable Chemistry for the Environment* 5 (Mar. 2024), p. 100056. DOI: 10.1016/j.scenv.2023.100056.

[8] **Antonios P. Sarikas**, George S. Fanourgakis, Emmanuel Tylianakis, Konstantinos Gkagkas, and George E. Froudakis. "Comparison of Energy-Based Machine Learning Descriptors for Gas Adsorption". In: *The Journal of Physical Chemistry C* (Oct. 2023). DOI: 10.1021/acs.jpcc.3c04223.

[9] **Antonios P. Sarikas**, George S. Fanourgakis, and George E. Froudakis. "Metal-organic frameworks in the age of machine learning". In: *Reticular Chemistry and Applications*. De Gruyter, Jan. 2023. DOI: 10.1515/9781501524721.

Peer-review activities

2026 Reviewer for *npj Artificial Intelligence*

2026 Reviewer for *npj Computational Materials*

2025 Reviewer for *Scientific Reports*

2025 Reviewer for *Communications Chemistry*

2024 Reviewer for *Journal of Cheminformatics*

</> Software

AIdsorb python package for deep learning on porous materials

- Homepage
- Documentation

POLiPaiR Pocket-Ligand Pair Ranker A web app for evalutating pocket-ligand fitness via machine learning

- Click to open the app

AIdsorb-online A web app for predicting properties of porous materials via deep learning

- Click to open the app

moXελ python package for parallel calculation of energy voxels

- Homepage
- Documentation

RetNeXt A pretrained model for transfer learning across the MOF adsorption space

- Homepage

RetNet A 3D ConvNet for Reticular Chemistry

- Homepage

Presentations

2026 3C Summer School V3, Heraklion
2025 Memorial seminar in honor of Dr. Chatzimarinaki, Heraklion
2025 Young Scientist Symposium (EuroMOF2025), Heraklion
2025 EuroMOF2025, Heraklion
2025 FEMS Masters Thesis Award, Granada
2025 FEMS EUROMAT 2025, Granada
2025 3C Summer School V2, Heraklion
2025 HSSTCM Masters Thesis Award, Online
2024 COST Action EU4MOFs, Istanbul
2023 Memorial seminar in honor of Dr. Chatzimarinaki, Heraklion
2023 1st Mediterranean Conference on Porous Materials, Rethymnon
2022 Memorial seminar in honor of Dr. Chatzimarinaki, Heraklion
2022 XXXVI Pan-Hellenic conference on Solid-State Physics and Materials Science, Heraklion

Workshops

2024 Computational Materials Science, University of Thessaly, Online
2023 Computational Materials Science, National Kapodistrinan University of Athens, Online
2022 Computational Materials Science, University of Ioannina, Online

2021 Computational Materials Science, University of Patras, Online

2021 DCMS Materials 4.0 Summer School, TU Dresden, Online

Computer skills

Operating Systems: Fedora Linux, Ubuntu, Windows

Programming Languages: Python

Markup Languages: LaTeX, Markdown

Other Tools: Git, Github, Sphinx, Vim, MSOffice

Working experience

2026 Lecturer, Machine Learning session, 3C Summer School V3, Department of Chemistry, University of Crete

2025 Lecturer, Machine Learning session, 3C Summer School V2, Department of Chemistry, University of Crete

2025 Coordinator for 3C Summer School V2, Department of Chemistry, University of Crete

2023, 2025 Teaching assistant on the Laboratory of Physical-Chemistry II, Department of Chemistry, University of Crete

2023, 2024 Teaching assistant on the Laboratory of Physical-Chemistry I, Department of Chemistry, University of Crete

Languages

English: Proficiency, University of Michigan

Greek: Native language